Pre-work · Cell Anatomy

Cell Anatomy. Read the Part A chapter first, then work this in order. You come to class ready to use this, not to hear it for the first time.

Module 1 · 2 of 4 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the cell chapter into mini visuals. Eight chunks, eight small visuals, each in the shape that fits it.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 **HUB**

The generalized cell

One hub, three spokes: plasma membrane, nucleus, cytoplasm. Hang one job off each spoke. This is the frame everything else attaches to.

2 **GRID**

Permeability

Semipermeable, permeable, impermeable down the side. Definition and one example across the top. Three rows, and the examples must be yours.

3 **HUB**

Inside the nucleus

Nucleus at the center. Spoke to envelope, pores, chromatin, chromosomes, nucleoplasm, nucleolus. One job on each spoke.

4 **SPLIT**

Cytoplasm versus cytosol

One split. The trap is that one contains the other. Write the containing relationship on the branch, not just two definitions.

5 **CHAIN**

The secretory pathway

Nucleus, arrow, rough ER, arrow, vesicle, arrow, Golgi, arrow, vesicle, arrow, plasma membrane. Then write what happens to the protein at each arrow.

6 **GRID**

Rough ER against smooth ER

The two down the side. Structure, function, and what it sends onward across the top.

7 **HUB**

The cytoskeleton

Three spokes: microfilaments, intermediate filaments, microtubules. Protein on one line, job on the next. Circle the one that builds cilia.

8 **SPLIT**

9 plus 0 against 9 plus 2

Centrioles on one branch, cilia and flagella on the other. Write the arrangement and, underneath, the one word that separates them: movement.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

The generalized cell, labelled

BRAIN DUMP FIRST

Two minutes. Every organelle you can name and one job for each.

Draw one cell large enough to label. Include the plasma membrane, nucleus with nucleolus, rough and smooth ER, Golgi, mitochondria, lysosomes, peroxisomes, ribosomes, cytoskeleton, centrioles. Label every one. Do not draw a diagram you have seen, draw the one you remember.

In your notebook, head the page **M1-4 The generalized cell, labelled** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can point at any organelle on your drawing and name what fails if it stops working.

DRAWING 2

The plasma membrane, close up

BRAIN DUMP FIRST

One minute. Every component of the membrane and where it sits.

Draw a section of bilayer wide enough to show detail. Mark the hydrophilic heads and hydrophobic tails and which way each faces. Add an integral protein spanning the layer, a peripheral protein on the surface, cholesterol between the tails, and a glycoprotein with its sugar chain facing out. Label which side is extracellular.

In your notebook, head the page **M1-5 The plasma membrane, close up** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain why a lipid-soluble drug crosses this and a charged ion does not, using only your drawing.

DRAWING 3

Trace the pathway on your cell

BRAIN DUMP FIRST

One minute. The route a secreted protein takes, in order.

Go back to your first drawing. In a second color, draw the route a secreted protein takes from the gene to the outside of the cell. Number each step. Mark where the protein is made, where it is modified, and how it moves between stations.

In your notebook, head the page **M1–6 Trace the pathway on your cell** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can tell the story out loud without looking, and your numbers are in the right order.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A drug destroys the enzymes inside lysosomes. Predict what accumulates and where.

Answer. Undigested macromolecules and worn organelles build up inside the cell.

Reasoning. Work from structure to consequence. The lysosome is the digestive organelle, so its job is breaking down macromolecules, damaged organelles, and pathogens. Remove the enzymes and nothing else takes over that job, so the material has nowhere to go and stays inside the cell. Cells with the highest turnover show it first. This is the mechanism behind Tay-Sachs, where one missing lysosomal enzyme lets lipid accumulate in nerve cells until the cell fails.

Set A · Structure to job

1. The organelle that assembles ribosomal subunits is the ________________________.
2. A cell that secretes large amounts of protein would be rich in ________________________.
3. A cell that detoxifies drugs would be rich in ________________________.
4. The organelle with its own DNA and ribosomes is the ________________________.
5. Name one cell type that is multinucleated and one that is anucleated.

   ____

Set B · Predict the failure

1. A mutation stops the cell producing glycoproteins. Predict two things the cell can no longer do well, and say why.
2. A cell cannot make enough ATP. Name the organelle involved, and name the three tissues that would show it first. Explain your choice.
3. The cilia lining the airway stop moving. Predict the clinical picture, and say which microtubule arrangement has failed.
4. Cholesterol is removed from the membrane. Predict what happens to the membrane as temperature changes.

Set C · Tell it apart

1. A student says cytoplasm and cytosol are the same thing. Correct them in one sentence.
2. Free ribosomes and bound ribosomes make different proteins. Say what decides which is used.
3. Chromatin and chromosomes are the same material. Say what changes between them and when.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M1-1

The generalized cell

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M1-2

Permeability

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M1-3

Inside the nucleus

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M1-4

Cytoplasm versus cytosol

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M1-5

The secretory pathway

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

M1-6

Rough ER against smooth ER

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M1-7

The cytoskeleton

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M1-8

9 plus 0 against 9 plus 2

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.